

## ACHISAY, Kazakhstan

WMO: 381960

Lat: 43.550N

Lon: 68.900E

Elev: 822

StdP: 91.83

Time Zone: 6.00 (E06)

Period: 01-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |       | Humidification DP/MCDB and HR |     |      |       |     |      | Coldest Month WS/MCDB |       |      |       | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-------|-------------------------------|-----|------|-------|-----|------|-----------------------|-------|------|-------|-----------------------|------|-------|
|               |            |       | 99.6%                         |     |      | 99%   |     |      | 0.4%                  |       | 1%   |       |                       |      |       |
|               | 99.6%      | 99%   | DP                            | HR  | MCDB | DP    | HR  | MCDB | WS                    | MCDB  | WS   | MCDB  | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)   | (d)                           | (e) | (f)  | (g)   | (h) | (i)  | (j)                   | (k)   | (l)  | (m)   | (n)                   | (o)  | (p)   |
| 1             | -14.9      | -12.6 | -23.8                         | 0.5 | -9.9 | -20.6 | 0.7 | -8.1 | 16.2                  | -12.2 | 15.2 | -12.6 | 2.8                   | 320  | 0.435 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 8.9                    | 34.4            | 16.6 | 32.9 | 16.2 | 31.5 | 15.7 | 17.6                | 30.3 | 17.1 | 29.8 | 16.5 | 29.0 | 2.3                  | 0    |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |     |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|-----|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |     |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR  | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h) | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 13.8                            | 10.8 | 20.2 | 12.6 | 10.1 | 20.0 | 11.5 | 9.4 | 19.8 | 53.0          | 30.2 | 51.0 | 29.7 | 49.2 | 29.1 |                   |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |     |     |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|-----|-----|
|                   |      |     | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |     |     |
| 1%                | 2.5% | 5%  | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  | DB  | WB  |
| (a)               | (b)  | (c) | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |     |     |
| (4) 7.4           | 5.7  | 4.5 | -17.3                      | 37.2 | 3.2                | 1.2 | -19.6                                              | 38.1 | -21.5      | 38.8 | -23.3      | 39.5 | -25.6      | 40.4 |     |     |
| (5)               |      |     | -17.8                      | 19.3 | 3.1                | 1.0 | -20.0                                              | 20.0 | -21.9      | 20.6 | -23.6      | 21.2 | -25.9      | 21.9 | (4) | (5) |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual      | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |      |      |
|------------------------------------------------------------------------|-------------|-------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|------|
|                                                                        |             | (d)         | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |      |      |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 11.6        | -1.8  | -1.3  | 5.7   | 11.5  | 18.0  | 23.2  | 25.7  | 24.4  | 18.5  | 11.1  | 4.3   | -0.7  | (6)  | (7)  |
|                                                                        | DBStd       | 10.97       | 5.01  | 5.68  | 5.39  | 5.16  | 4.07  | 3.55  | 3.47  | 3.72  | 4.59  | 5.31  | 5.75  | 5.48  | (7)  | (8)  |
|                                                                        | HDD10.0     | 1444        | 366   | 316   | 152   | 42    | 2     | 0     | 0     | 0     | 2     | 50    | 183   | 332   | (8)  | (9)  |
|                                                                        | HDD18.3     | 3129        | 624   | 549   | 392   | 209   | 58    | 5     | 1     | 3     | 53    | 228   | 419   | 589   | (9)  | (10) |
|                                                                        | CDD10.0     | 2038        | 1     | 1     | 18    | 88    | 248   | 395   | 486   | 445   | 257   | 84    | 13    | 1     | (10) | (11) |
|                                                                        | CDD18.3     | 682         | 0     | 0     | 0     | 5     | 46    | 150   | 228   | 189   | 59    | 4     | 0     | 0     | (11) | (12) |
|                                                                        | CDH23.3     | 6717        | 0     | 0     | 1     | 28    | 303   | 1472  | 2565  | 1907  | 421   | 21    | 0     | 0     | (12) | (13) |
|                                                                        | CDH26.7     | 2579        | 0     | 0     | 0     | 2     | 57    | 532   | 1110  | 756   | 119   | 3     | 0     | 0     | (13) | (14) |
| (14) Wind                                                              | WSAvg       | 2.1         | 1.8   | 2.0   | 2.0   | 2.4   | 2.4   | 2.1   | 2.2   | 2.6   | 2.5   | 1.9   | 1.7   | 1.7   | (14) | (15) |
| (15) Precipitation                                                     | PrecAvg     | 397         | 47    | 48    | 56    | 48    | 39    | 13    | 12    | 4     | 5     | 24    | 48    | 55    | (15) | (16) |
|                                                                        | PrecMax     | 731         | 108   | 110   | 115   | 143   | 125   | 49    | 53    | 19    | 28    | 69    | 161   | 262   | (16) | (17) |
|                                                                        | PrecMin     | 158         | 0     | 0     | 15    | 0     | 8     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | (17) | (18) |
|                                                                        | PrecStd     | 126         | 26    | 27    | 32    | 37    | 31    | 14    | 14    | 6     | 7     | 23    | 40    | 50    | (18) | (19) |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB          | 12.1  | 14.6  | 20.8  | 26.2  | 30.3  | 34.7  | 37.0  | 36.0  | 32.1  | 26.0  | 20.4  | 13.3  | (19) | (20) |
|                                                                        |             | MCWB        | 5.2   | 8.0   | 11.2  | 14.7  | 15.3  | 17.0  | 17.4  | 16.9  | 15.0  | 12.9  | 11.2  | 5.9   | (20) | (21) |
|                                                                        | 2%          | DB          | 8.5   | 10.6  | 17.7  | 23.6  | 28.2  | 32.8  | 34.9  | 34.0  | 29.7  | 22.8  | 16.8  | 10.2  | (21) | (22) |
|                                                                        |             | MCWB        | 3.7   | 6.0   | 9.9   | 13.7  | 14.7  | 16.3  | 16.8  | 16.4  | 14.1  | 11.1  | 8.9   | 4.4   | (22) | (23) |
|                                                                        | 5%          | DB          | 6.4   | 8.1   | 15.2  | 21.3  | 26.5  | 31.3  | 33.2  | 32.3  | 27.8  | 20.6  | 14.3  | 8.2   | (23) | (24) |
|                                                                        |             | MCWB        | 2.3   | 4.3   | 8.7   | 12.4  | 13.9  | 15.8  | 16.3  | 15.8  | 13.3  | 10.2  | 8.0   | 3.1   | (24) | (25) |
|                                                                        | 10%         | DB          | 4.5   | 6.1   | 13.1  | 19.1  | 24.9  | 29.7  | 31.8  | 30.7  | 25.7  | 18.3  | 11.9  | 6.4   | (25) | (26) |
|                                                                        |             | MCWB        | 1.1   | 3.0   | 8.0   | 11.5  | 13.3  | 15.4  | 15.9  | 15.2  | 12.6  | 9.3   | 7.0   | 2.3   | (26) | (27) |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB          | 6.3   | 8.5   | 12.7  | 16.2  | 16.6  | 18.5  | 18.7  | 17.8  | 15.7  | 13.8  | 11.7  | 7.1   | (27) | (28) |
|                                                                        |             | MCDB        | 10.2  | 12.4  | 16.3  | 23.0  | 26.4  | 28.9  | 31.8  | 31.6  | 30.7  | 23.2  | 17.3  | 11.4  | (28) | (29) |
|                                                                        | 2%          | WB          | 4.2   | 6.8   | 11.1  | 14.7  | 15.8  | 17.5  | 17.8  | 17.0  | 14.5  | 12.2  | 10.6  | 5.3   | (29) | (30) |
|                                                                        |             | MCDB        | 7.5   | 9.9   | 16.1  | 20.6  | 25.2  | 29.1  | 30.6  | 31.3  | 27.8  | 20.2  | 15.1  | 8.1   | (30) | (31) |
|                                                                        | 5%          | WB          | 3.0   | 5.0   | 9.7   | 13.5  | 15.0  | 16.8  | 17.2  | 16.4  | 13.7  | 11.1  | 8.9   | 4.0   | (31) | (32) |
|                                                                        |             | MCDB        | 5.4   | 7.3   | 14.5  | 19.7  | 23.7  | 28.4  | 30.2  | 30.5  | 26.5  | 18.6  | 13.5  | 7.3   | (32) | (33) |
|                                                                        | 10%         | WB          | 1.8   | 3.3   | 8.6   | 12.3  | 14.3  | 16.2  | 16.6  | 15.7  | 12.8  | 10.0  | 7.2   | 2.7   | (33) | (34) |
|                                                                        |             | MCDB        | 4.1   | 5.8   | 12.5  | 18.1  | 22.6  | 27.6  | 29.5  | 29.4  | 24.9  | 17.3  | 11.2  | 6.2   | (34) | (35) |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR        | 6.2   | 6.7   | 7.5   | 7.7   | 8.8   | 9.4   | 8.9   | 8.5   | 8.1   | 8.1   | 7.2   | 6.8   | (35) | (36) |
|                                                                        |             | MCDBR       | 7.8   | 8.5   | 10.3  | 10.3  | 10.4  | 10.7  | 9.6   | 9.9   | 10.0  | 10.6  | 9.6   | 8.3   | (36) | (37) |
|                                                                        | 5% WB       | MCWBR       | 5.2   | 5.4   | 5.7   | 5.4   | 4.4   | 4.0   | 3.6   | 3.9   | 4.3   | 5.2   | 5.3   | 5.3   | (37) | (38) |
|                                                                        |             | MCWBR       | 6.5   | 7.6   | 9.1   | 8.7   | 9.8   | 10.2  | 9.7   | 9.5   | 9.8   | 9.7   | 8.6   | 7.5   | (38) | (39) |
| (39) Clear-Sky Solar Irradiance                                        | taub        | taub        | 0.331 | 0.367 | 0.424 | 0.453 | 0.441 | 0.425 | 0.413 | 0.379 | 0.356 | 0.381 | 0.374 | 0.327 | (39) | (40) |
|                                                                        |             | taud        | 2.057 | 2.031 | 1.987 | 1.971 | 2.055 | 2.132 | 2.176 | 2.263 | 2.286 | 2.122 | 2.081 | 2.091 | (40) | (41) |
|                                                                        | Ebn at Noon | Ebn at Noon | 755   | 797   | 799   | 811   | 835   | 848   | 854   | 874   | 866   | 773   | 697   | 719   | (41) | (42) |
|                                                                        |             | Edh at Noon | 107   | 131   | 157   | 173   | 163   | 152   | 144   | 127   | 114   | 118   | 104   | 94    | (42) | (43) |
| (43) All-Sky Solar Radiation                                           | RadAvg      | RadAvg      | 1.62  | 2.60  | 3.79  | 5.39  | 6.88  | 7.64  | 7.50  | 6.80  | 5.33  | 3.38  | 1.93  | 1.36  | (43) | (44) |
|                                                                        | RadStd      | RadStd      | 0.23  | 0.27  | 0.29  | 0.53  | 0.41  | 0.33  | 0.33  | 0.22  | 0.22  | 0.26  | 0.20  | 0.15  | (44) | (45) |

## Historical Trends

|      |                        | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|------|------------------------|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|      |                        |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
| (46) | Station Only           | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)     | (l)     | (m)     |
| (47) | Regional (6 neighbors) | N/A   | N/A     | N/A    | +0.97   | N/A   | N/A   | N/A         | N/A     | +183    | +133    |

Nomenclature: See separate page